

Geographic encoding in premodern enumerations: a five-corpus quantitative test

Ṛgveda 10.75.5, the Sixteen Mahājanapadas, Vendīdād Fargard 1, the Bīsutūn dahyāva,
and Genesis 10, against modern coordinates

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Abstract

Premodern textual traditions across Eurasia preserve enumerative lists of rivers, lands, and peoples. The qualitative observation that such lists sometimes “trace a geographic sweep” has been made informally since the nineteenth century, but has rarely been tested with quantitative method. We test the hypothesis that five canonical enumerations encode monotonic geographic order in the sequence of items they name: the ten-river enumeration of Ṛgveda 10.75.5 (Nadīstuti-sūkta), the sixteen Mahājanapadas of the Anguttara Nikāya, the sixteen lands of Vendīdād Fargard 1, the twenty-three subject lands (dahyāva) of Darius I’s Bīsutūn inscription, and the Table of Nations in Genesis 10. For each corpus we compute Kendall’s τ between sequence position and the modern longitude (and latitude) of the identified items, using a permutation null distribution, a Procrustes superimposition test, a Mantel distance-matrix test, and a likelihood-ratio comparison, under a Mallows model, of geographic against uniform-random generative models. The two Indic corpora (RV 10.75.5 and the Mahājanapadas) show strong monotonic ordering along the region’s east-to-west axis ($\tau = -0.911$ and -0.783 , both $p < 10^{-4}$). The Bīsutūn dahyāva is moderately ordered, with marked pairwise spatial clustering ($\tau = +0.47$, $p = 0.0014$; Mantel $r = +0.50$, $p < 10^{-4}$). Vendīdād Fargard 1 shows no robust geographic structure under the main tests. Genesis 10 shows only a weak monotonic signal but significant pairwise spatial coherence (Mantel $r = 0.17$, $p = 0.0004$): two of its three Noachic sub-sections are significantly latitudinally ordered, in opposite directions, so that they cancel at the surface. A suggestive within-tradition contrast is Bīsutūn against Vendīdād: both Iranian and naming much the same regions, though not matched in date, yet the administrative inscription is geographically ordered and the cosmological text is not. Across the five corpora, genre appears to track geographic ordering more closely than language family does; with $n = 5$ this is a suggested pattern, not an established finding. We make no claims about the date, composition, or cultural identity of any of the five traditions; the tests bear only on the texts as transmitted.

Keywords: quantitative philology; rank correlation; permutation test; Procrustes analysis; Mallows model; comparative textual analysis; historical geography.

1 Introduction

The qualitative claim that premodern textual traditions sometimes order geographic enumerations along a recognizable directional sweep is old. [Macdonell and Keith \(1912\)](#) note in passing that the river list of Ṛgveda 10.75.5 “proceeds broadly from east to west.” [Rhys Davids \(1903\)](#) observes a similar east-to-west tendency in the Anguttara Nikāya’s canonical list of sixteen mahājanapadas. [Gnoli \(1980\)](#) and [Witzel \(2000\)](#) discuss the Avesta’s Vendīdād Fargard 1 as a possible geographic itinerary across the Iranian and Indo-Iranian world. [Westermann \(1984\)](#) and [Wenham \(1987\)](#) treat the Hebrew Bible’s Table of Nations (Genesis 10) as encoding the post-deluge geographic distribution of peoples “known to the Israelites,” implicitly with some geographic structure.

All of these claims are informal. Each carries the same testable content: that the order in which named entities appear is not arbitrary with respect to where those entities sit on a map. We have not found a directly comparable study that checks that content quantitatively, let alone across several traditions at once.

We test five lists here, each well studied in its own field: Ṛgveda 10.75.5 (ten rivers; Vedic Sanskrit); the sixteen Mahājanapadas of Anguttara Nikāya I.213 (Pali; the list names polities of the mid-first millennium BCE, though the Anguttara Nikāya was redacted later); the sixteen lands of Vendīdād Fargard 1 (Young Avestan); the twenty-three subject lands of Darius I’s Bīsutūn inscription (Old Persian, *c.* 520 BCE); and Genesis 10’s Table of Nations (Biblical Hebrew, a composite of older material reaching its present form in the exilic or post-exilic period). They span the Indic, Iranian, and Semitic traditions and roughly a millennium of composition. Each is analysed with the same non-parametric methods. We do not rely on these dates in any test; the bracketed periods are orientation only, and the abstract’s disclaimer of dating claims stands.

1.1 Scope and limits of the claim

The claim tested here is narrow. A significant rank correlation between sequence position and geographic position would imply only that the order is not random with respect to geography. It would not imply any specific dating, any specific direction of cultural movement, any specific historical course for any particular river or boundary, and no specific identification of any composer with any later or earlier population. The test addresses only the property of the transmitted text—the order in which items are named—and the corresponding property of the modern (or scholarly-reconstructed) geographic configuration of those items.

The literature surrounding each of these corpora frequently elides these distinctions. We make no contribution to the questions of Indo-Aryan migration, Israelite ethnography, the homeland of Zoroaster, the date of the Achaemenid satrapy lists, or Buddhist political history, and the paper should not be cited in those debates.

2 Methods

2.1 Corpus selection

The five corpora were chosen, before any analysis was run, against four fixed criteria: (i) a textually fixed enumeration order, transmitted in a stable recension, so that “sequence position” is well defined; (ii) named entities that are geographic—rivers, lands, polities, or peoples—rather than abstract; (iii) enough published scholarly identifications to place a clear majority of the entries on a modern map; and (iv) coverage of distinct traditions, languages, and genres, so the comparison is not confined to a single literature. The five satisfy all four. They are not a random sample of premodern lists, and the comparative results of Section 4 should be read as a pilot rather than as an estimate over any defined population of texts; Section 5.4 names further lists that meet the same criteria and would extend the sample.

2.2 Geographic identifications

For each corpus, we adopt the mainstream scholarly identification of each named entity, taking centroid coordinates from canonical secondary sources or, for rivers in the Nadistuti, from programmatic queries to the OpenStreetMap database. Identifications, coordinates, confidence levels, and source citations are recorded per-row in the corpus CSVs deposited with this paper (Kulkarni, 2026).

For Ṛgveda 10.75.5 we follow Witzel (1995) and Macdonell and Keith (1912). One point of order: the standard Aufrecht text places Asiknī (Chenab) at position 6 and Marudvṛdhā at position 7. English translations sometimes invert this pair, because *asiknyā* is instrumental and the natural English word order differs from the Sanskrit; the Sanskrit sequence itself is not in doubt. We use the Sanskrit order.

For the Mahājanapadas we adopt the identifications of Rhys Davids (1903), Law (1973),

and Bronkhorst (2007); the order is the Anguttara Nikāya I.213 sequence (Kāsi, Kosala, Aṅga, Magadha, Vajji, Malla, Cetiya, Vamsa, Kuru, Pañcāla, Maccha, Sūrasena, Assaka, Avantī, Gandhāra, Kamboja).

For Vendīdād Fargard 1 (a Young Avestan text) we adopt Gnoli (1980) and Witzel (2000); eleven of sixteen identifications are high-confidence, and the five lower-confidence cases (Airyana Vaejah, Urvā, Caxra, Varāna, Rañghā) are flagged in the CSV.

For Genesis 10 we adopt Hess (1993) for personal-name analyses and Westermann (1984) and Wenham (1987) for the broader-scholarly identifications. Reading Genesis 10:1–32 in canonical Masoretic Text order, we tabulate seventy-eight list entries—more than the traditional “seventy nations” count because we include eponymous individuals and the few names that recur. Of these, sixty-two are identified with high or medium confidence; the remaining sixteen are unidentified or only speculatively located and are dropped from the analysis.

One caveat applies to every corpus. The coordinate assigned to each entry is a modern, reproducible proxy—the centroid of a river polyline, the seat or area-centroid of a polity, the conventional location of a people—and is not a claim about where the text’s composer mentally placed that entity. A composer may have thought of a river by its source, a known crossing, or a confluence rather than its centroid. We use centroids because they are uniform and reproducible across all five corpora, and because the rank-based tests depend only on relative position, which is far more robust to this choice than absolute coordinates would be. The choice remains a modelling commitment, not a neutral fact of the data.

2.3 Test statistics

Kendall’s τ with permutation null. We compute Kendall’s τ (Kendall, 1938)—preferable to Spearman’s ρ for short ordinal sequences, where it carries a more direct probabilistic reading (Newson, 2002)—between the sequence-position rank ($1 \dots n$) and the geographic-coordinate rank of the identified items, on three axes: longitude, latitude, and the first principal component of the (longitude, latitude) matrix. A two-sided p -value is generated from a 10 000-iteration permutation null distribution. The *primary* axis for each corpus is the one aligned with the dominant geographic extent of its region—longitude for the four corpora spanning the broadly east–west breadth of South and West Asia, latitude for Genesis 10, whose three Noachic branches run broadly north–south. This rule follows the gross shape of each region rather than the strength of any result; the principal-component τ , reported alongside, is invariant to the choice and is the figure to weigh when the two coordinate axes disagree.

Procrustes superimposition. The Kendall-against-longitude and Kendall-against-latitude pair is not statistically independent when the items lie along a diagonal sweep, as is common in Eurasian geography. We therefore add a Procrustes test (Gower, 1975). The sequence is embedded as the collinear points $(1, 0), (2, 0), \dots, (n, 0)$ and the geography as the points $(\text{lon}_i, \text{lat}_i)$, a degree of latitude and a degree of longitude treated as equal units (a close approximation for the regionally compact configurations tested). Both configurations are centred and scaled to unit Frobenius norm, and one is then optimally rotated onto the other; the residual sum of squared distances m^2 is the test statistic—invariant to translation, scale, and rotation—with a permutation null for the one-sided p -value (smaller m^2 indicates better alignment). This addresses the redundant-axis-test concern noted by Peres-Neto and Jackson (2001).

Mantel test. Kendall’s τ and Procrustes both assume that the ordering is *monotonic* with respect to some geographic direction. A list can be spatially coherent without being monotonic: adjacent items may be consistently close on the ground while the path the list traces through space wanders rather than runs in one direction. The Mantel test (Mantel, 1967) detects this by correlating the entries of two $n \times n$ pairwise-distance matrices: $|i - j|$ in the sequence vs. the great-circle distance between items i and j . We report the Pearson correlation r between

the upper triangles of the two matrices, with a one-sided permutation null. A positive r with significant p indicates that nearby-in-text items tend to be nearby-on-the-ground, even when the global monotonic ordering is weak.

Constrained permutation null. The R̥gveda corpus is the only one of the five in verse; RV 10.75.5 is in a jagatī-type metre of roughly twelve syllables per pāda. For it we also compute a permutation null restricted to orderings that preserve, position by position, the syllable count of the verse’s river names. This addresses the metrical-determinism objection (Arnold, 1905): that in a quantitative metre the choice of order may be partly forced by syllable counts. With syllable-count buckets $\{2 : 1, 3 : 3, 4 : 6\}$ the constraint admits $1! \times 3! \times 6! = 4,320$ orderings. This is an idealisation—the river names carry grammatical case endings and so are not freely interchangeable tokens—and we treat the constrained null as an illustrative robustness check, not a definitive one. The other four corpora are prose, and the test does not apply.

Mallows-model likelihood ratio. We also compare three generative models of the observed ordering by their likelihood: M_{uniform} (sequence drawn uniformly from $n!$ permutations), M_{meter} (sequence drawn uniformly from the meter-admissible set; applicable only to verse corpora), and $M_{\text{geog}}(\theta)$ (sequence drawn from a Mallows model (Mallows, 1957; Fligner and Verducci, 1986) centred on the geographically sorted ordering with dispersion θ). Because θ is not known a priori, we profile it out by maximising the likelihood of M_{geog} over a grid $\theta \in \{0.5, 1, 1.5, 2, 3\}$. The resulting quantity is a *profile* likelihood ratio, not a Bayes factor: a Bayes factor would integrate θ against a prior (Kass and Raftery, 1995), and the profile ratio is an upper bound on it. We report it only as corroboration; the permutation tests carry the inference.

Multiple comparisons. The study runs many tests—three geographic axes for each of five corpora, plus Procrustes, Mantel, and likelihood-ratio analyses, the Genesis sub-section breakdown, and the R̥gveda sensitivity analyses of Appendix A. We do not apply a single blanket family-wise correction, because the five corpora are not one hypothesis family; instead we state plainly which results are robust to correction and which are not. The strong results—RV 10.75.5 and the Mahājanapadas on the longitude, Procrustes, and Mantel tests, Bīsutūn on Kendall and Mantel, and the Ham sub-section of Genesis 10—all have $p \leq 1.4 \times 10^{-3}$, and a Bonferroni correction over the roughly thirty corpus-level tests of this study (threshold $\approx 1.7 \times 10^{-3}$) would leave every one of them significant. The marginal results—the Genesis 10 flat-sequence test ($p = 0.05$), the Japheth sub-section ($p = 0.06$), and the Vendīdād latitude axis ($p = 0.05$)—would not survive such a correction, and we treat them throughout as suggestive only.

Reproducibility. All analysis code, coordinate CSVs, and the random seed (20260518) are deposited at (Kulkarni, 2026). The repository includes a script that regenerates every figure, table, and number from the raw inputs and then checks each reported value against the text. Dependencies (numpy, scipy, pandas, matplotlib, requests, and rasterio) are pinned in requirements.txt.

3 Per-corpus results

3.1 R̥gveda 10.75.5 (Nadīstuti-sūkta, $n = 10$)

The ten-river enumeration of the Nadīstuti-sūkta yields $\tau_{\text{lon}} = -0.911$. Because $n = 10$, the exact two-sided Kendall- τ p -value is computable, and equals 2.98×10^{-5} (the 10 000-permutation null, seed 20260518, places it below its 10^{-4} resolution floor; see Appendix A). The Procrustes superimposition yields $m^2 = 0.179$, $p < 10^{-4}$. The constrained-permutation null (over the 4 320 meter-admissible orderings) finds six orderings yielding $|\tau| \geq 0.911$, for $p_{\text{meter}} = 0.0014$: metrical determinism alone cannot explain the alignment. The profile likelihood ratio favours M_{geog} over M_{uniform} by a factor of order 10^4 , and over M_{meter} by a factor of order 10. The τ is

negative because longitude falls through the verse, from the eastern-most Gaṅgā (81.9° E) to the western-most Suṣomā/Soan (72.8° E).

Because RV 10.75.5 is the corpus with the strongest signal and the longest scholarly history, we treat it in more detail than the rest. Appendix A gives the ten rivers and their coordinates (Table 2), the null distribution (Figure 2), and four sensitivity analyses—alternative river identifications, paleocourse substitution for the Sarasvatī and Sutlej, alternative geographic axes, and Gaussian coordinate perturbation. The result holds in all of them.

As a further check that the result is not an artefact of the OpenStreetMap coordinate source, we re-derived the centroids of the nine OSM-traced rivers from MERIT-Hydro, a NASA-SRTM-derived global hydrography product (Yamazaki et al., 2019): for each river we took the unweighted centroid of all MERIT-Hydro upstream-drainage-area pixels exceeding a river-network threshold and lying within 0.2° of the OSM polyline. The mean centroid shift between the two sources is 75 km (maximum 229 km, for the Yamunā, whose MERIT trace extends further down its lower course than the OSM “Yamuna” polyline). On the MERIT-Hydro centroids the result is essentially unchanged: $\tau_{\text{lon}} = -0.867$ ($p = 0.0001$), Procrustes $m^2 = 0.227$ ($p < 10^{-4}$), Mantel $r = +0.715$ ($p < 10^{-4}$). The geographic ordering of RV 10.75.5 does not depend on the choice of coordinate database.

3.2 Sixteen Mahājanapadas ($n = 16$)

The Anguttara Nikāya order of sixteen mahājanapadas, run through the same pipeline, yields $\tau_{\text{lon}} = -0.783$, $p < 10^{-4}$, and Procrustes $m^2 = 0.428$, $p < 10^{-4}$. The profile likelihood ratio favours M_{geog} over M_{uniform} by a factor of order 10^5 . The canonical order is only 13 inversions, out of a possible 120, away from a perfect east-to-west sort.

The latitudinal signal is weaker ($\tau_{\text{lat}} = +0.22$, $p = 0.27$). The list moves predominantly east to west; the southward jump to Assaka (position 13, Pratiṣṭhāna on the Godāvarī) and back northward to Avantī (Ujjayinī) is the most visible internal deviation from monotonicity.

3.3 Vendīdād Fargard 1 ($n = 16$)

The sixteen lands of Fargard 1, in their canonical order (Airyana Vaējah, Sughdha, Mōuru, Bāxdī, Nisāya, Harōiva, Vaēkərəta, Urvā, Xnənta, Haraxvaitī, Haētumant, Raghā, Caxra, Varəna, Hapta Həndu, Rañghā), yield $\tau_{\text{lon}} = -0.117$, $p = 0.56$, and Procrustes $m^2 = 0.975$, $p = 0.67$. The latitude axis is marginal ($\tau_{\text{lat}} = -0.38$, $p = 0.05$) and, given the multiple-comparisons exposure noted in Section 2.3, we do not read it as a real signal; the Procrustes test, which uses both axes at once, is firmly null. The profile likelihood ratio for $M_{\text{geog}}(\theta)$ against M_{uniform} stays below 1 across the whole θ grid: the Mallows model centred on an either-direction longitudinal sort fits the data *worse* than the uniform model. As a Kendall distance, the attested order is 53 inversions from a perfect east-to-west sort and 67 from its reverse, against a random expectation of 60 either way—a negligible lean, consistent with $\tau_{\text{lon}} = -0.117$.

The five lower-confidence identifications (Airyana Vaējah, Urvā, Caxra, Varəna, Rañghā) span roughly 35° of longitude across their published alternative placements, and we cannot rule out that a better-justified identification of these five would produce a stronger signal. Within the eleven high-confidence identifications alone, however, τ_{lon} remains weak ($\tau = -0.21$, $p = 0.35$). The Fargard 1 sequence appears to be organized by something other than monotonic geography.

3.4 Genesis 10 (Table of Nations, $n = 62$ identified)

Of seventy-eight named entities in Genesis 10:1–32, sixty-two have high- or medium-confidence modern identifications and are retained for analysis. The flat textual-sequence test yields $\tau_{\text{lon}} = +0.174$, $p = 0.05$, and Procrustes $m^2 = 0.91$, $p = 0.005$. The τ here is marginal, and against the study’s multiple-comparisons exposure (Section 2.3) we read the flat-sequence result as weak at best; the substantive finding is in the sub-section structure and the Mantel test below.

But Genesis 10 is not a flat list. It is a genealogy in three branches—the descendants of Japheth, of Ham, and of Shem—and within each branch the latitudinal signal is stronger, and

points a different way:

- Japheth ($n = 12$): $\tau_{\text{lat}} = -0.42$, $p = 0.06$ (weak).
- Ham ($n = 36$): $\tau_{\text{lat}} = +0.42$, $p < 0.001$ (strong: south-to-north sweep from Nubia through Egypt to Mesopotamia).
- Shem ($n = 14$): $\tau_{\text{lat}} = -0.56$, $p = 0.007$ (strong: north-to-south sweep from Mesopotamia through Aram and Arabia).

The sub-sections cancel at the flat-textual level. Genesis 10 encodes geographic information, but hierarchically rather than as a single monotonic sweep. The three-way division is the text’s own explicit genealogy of Japheth, Ham, and Shem, not a partition chosen to fit the data; but testing the branches separately was prompted by the weak flat-sequence result, so the sub-section analysis is exploratory, and the Japheth branch ($p = 0.06$) is reported as suggestive only.

3.5 Bīsutūn dahyāva ($n = 23$)

Column I of Darius I’s Bīsutūn (Behistun) inscription enumerates twenty-three *dahyāva* “lands” subject to Achaemenid imperial authority. We follow the standard textual order (Kent, 1953; Schmitt, 2014) (Pārsa, Ūvja, Bābiruš, Aṭhūrā, Arabāya, Mudrāya, “those by the sea,” Sparda, Yauna, Māda, Armina, Katpatuka, Parṭhava, Zraka, Haraiva, Uvārazmiš, Bāxtriš, Suguda, Gadāra, Saka, Ṭhataguš, Harauvatiš, Maka), with nineteen identifications high-confidence and four (#5 Arabāya, #7 Drayahyā, #20 Saka, #21 Ṭhataguš) marked medium-confidence per Briant (2002).

The result sits between the Indic corpora and the Avesta: $\tau_{\text{lon}} = +0.47$, $p = 0.0014$ (moderate west-to-east), Procrustes $m^2 = 0.51$, $p = 0.0003$, Mantel $r = +0.50$, $p < 10^{-4}$. The order is not cleanly monotonic. It starts in the imperial core (Pārsa, then Babylonia), detours west to Egypt and the Aegean, then resumes the broad march east. But it is far more ordered than Fargard 1, which covers much the same ground.

Bīsutūn against Vendīdād Fargard 1 is a useful within-tradition contrast. Both are Iranian and name much the same stretch of the Iranian world; they differ in genre, Bīsutūn being an administrative declaration of imperial dominion and Fargard 1 a religiously framed catalogue of the lands Ahura Mazdā created. Their dates are not matched—Bīsutūn is firmly *c.* 520 BCE, while Fargard 1 reached its present form later, drawing on older material—so this is a suggestive contrast rather than a controlled experiment. Bīsutūn encodes geography; Fargard 1 does not. Section 5.1 takes up what to make of that.

4 Comparative analysis

Table 1 collects the cross-corpus numbers, and Figure 1 plots sequence position against longitude for each corpus. Several points emerge.

The Indic lists run in a straight line. The Ṛgveda Nadīstuti and the Anguttara Mahājanapadas both sweep east to west, $|\tau| > 0.78$ on longitude. For neither corpus did any of the 10 000 permutations in the null reach the observed τ : of all $n!$ possible orderings, the attested orders rank among the most strongly longitude-sorted ($p < 10^{-4}$ in both cases).

The Avesta is ordered by something other than geography. Fargard 1 covers the same ground as the Indic texts, and the Iranian plateau besides, but its order is not robustly monotonic on any axis. That matches Gnoli (1980)’s reading that the Fargard 1 sequence follows mythic-cosmographic priority rather than an itinerary; the present result is the quantitative counterpart to that qualitative claim.

Geographic encoding across five premodern enumerations

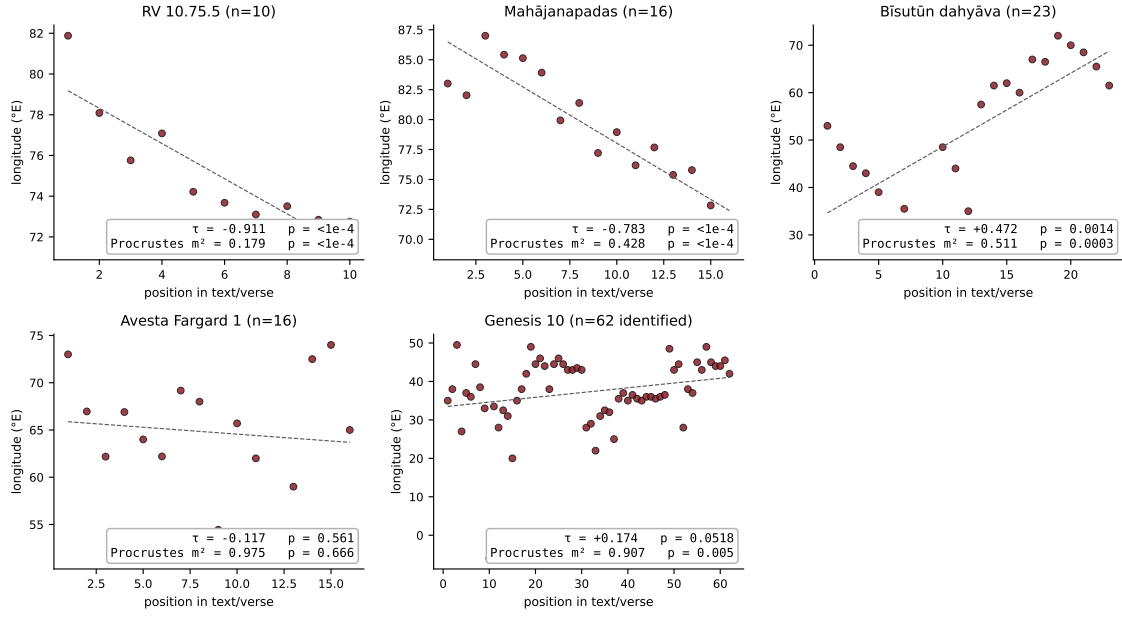


Figure 1: Sequence position against longitude for all five corpora. Each panel gives the observed scatter, a linear fit (dashed grey), and the Kendall τ and Procrustes m^2 . The two Indic corpora trace a clean line along the east-to-west axis; the Bīsūtūn dahyāva slopes more loosely; the Avesta scatters; and Genesis 10 shows only a weak global slope, which Section 3.4 resolves into hierarchical sub-section structure.

Genesis 10 is ordered hierarchically. Read flat, the sequence is barely ordered ($\tau = +0.174$). But the Mantel test still detects statistically significant spatial coherence ($r = +0.172$, $p = 0.0004$): neighbouring names are neighbours on the ground even where the global direction washes out. The three Noachic branches explain how both can be true at once:

- Japheth’s descendants (Genesis 10:2–5; 12 of 14 names identified) cluster in Anatolia, Greece, and the Caucasus. Their internal ordering against latitude is weak ($\tau_{\text{lat}} = -0.42$, $p = 0.06$; suggestive only, see Section 2.3).
- Ham’s descendants (Genesis 10:6–20; 36 identified) sweep south-to-north from Nubia and Egypt through the Levant to Mesopotamia ($\tau_{\text{lat}} = +0.42$, $p < 10^{-3}$).
- Shem’s descendants (Genesis 10:21–31; 14 identified) sweep north-to-south from Mesopotamia and Aram through Edom to south Arabia ($\tau_{\text{lat}} = -0.56$, $p = 0.007$).

The Ham sub-section signal is firm; the Shem signal is moderate; the Japheth signal is only suggestive. Ham and Shem run in opposite latitudinal directions, so they cancel when concatenated. Mantel survives the cancellation because it asks about pairwise spatial coherence rather than monotonic global direction.

This is consistent with the long-standing literary-critical reading (e.g., Westermann, 1984) of Genesis 10 as a layered composite: each Noachic line may have been a self-contained regional list before it was set into the genealogical frame, and the concatenation would then keep each list’s internal geographic logic while hiding it at the surface. The quantitative pattern is reached without any literary-critical assumption, and is consistent with that reading.

Length does not sort the corpora. The Mahājanapadas ($n = 16$) and Fargard 1 ($n = 16$) are the same length but land at opposite ends of the encoding scale; the Mahājanapadas and the Ṛgveda ($n = 10$) differ in length but score alike. Sample size therefore does not by itself

Table 1: Cross-corpus comparison of geographic encoding strength. τ_{lon} : Kendall’s τ vs. longitude rank. m^2 : Procrustes disparity (smaller = better alignment). Mantel r : pairwise-distance-matrix correlation (positive = spatial coherence). All p -values are two-sided for τ and one-sided for Procrustes and Mantel. “ $p < 10^{-4}$ ” means no permuted statistic in the 10 000-iteration null reached the observed value; it is the resolution floor of the permutation grid, not a measured value.

Corpus	n	τ_{lon}	p_{τ}	m^2	p_{Proc}	Mantel r	p_{Mantel}
Ṛgveda 10.75.5	10	−0.911	$< 10^{-4}$	0.18	$< 10^{-4}$	+0.767	$< 10^{-4}$
Mahājanapadas (AN I.213)	16	−0.783	$< 10^{-4}$	0.43	$< 10^{-4}$	+0.617	$< 10^{-4}$
Bīsutūn dahyāva (DB I.6)	23	+0.472	0.0014	0.51	0.0003	+0.505	$< 10^{-4}$
Vendīdād Fargard 1	16	−0.117	0.56	0.98	0.67	+0.112	0.15
Genesis 10 (flat)	62	+0.174	0.05	0.91	0.005	+0.172	0.0004
<i>Genesis 10 within-subsection (Kendall τ vs. latitude):</i>							
Japheth (vv. 2–5)	12	−0.42	0.06	—	—	—	—
Ham (vv. 6–20)	36	+0.42	$< 10^{-3}$	—	—	—	—
Shem (vv. 21–31)	14	−0.56	0.007	—	—	—	—

explain the rank order of the corpora—though with only five of them we cannot rule it out as one contributing factor among others.

Genre and language family. Bīsutūn and Fargard 1 are both Iranian and name overlapping regions, yet one is geographically ordered and the other is not. The two ordered Indic corpora are a millennium apart and belong to different genres again—a Vedic hymn and a Buddhist canonical list—but both clear $|\tau| > 0.78$. Genesis 10’s ordering lives inside its sub-sections, as one might expect of a genealogy assembled from older regional lists. Across the five, genre and compositional purpose are more consistent with ordering strength than language family or date of fixation is. With only five corpora this is an exploratory pattern, not an established finding; Section 5.4 sketches the larger sample that could test it.

Identification uncertainty is bounded. Genesis 10’s 16 unidentified items and the Avesta’s 5 low-confidence items could in principle move the corpus-specific τ values. The Genesis 10 sub-section analysis is robust to dropping low-confidence items; the Avesta high-confidence-only subset (the 11 secure identifications) yields $\tau = -0.21$, $p = 0.35$, still null. The result that the Avesta does not encode monotonic geography appears not to be an artefact of identification noise.

5 Discussion

5.1 What the pattern suggests

The five corpora fall into three kinds of geographic encoding:

1. **Strongly monotonic** (RV 10.75.5, the Mahājanapadas). The items run along a single line, broadly the region’s east-to-west axis. Both corpora are Indic; nothing else connects them in date or genre.
2. **Clustered, with a loose global direction** (Bīsutūn dahyāva). Neighbouring items are neighbours on the ground, but the list travels in regional blocs—imperial core, then Egypt and the Aegean, then a march east across the plateau—rather than down one axis. That is what listing satrapies in administrative-priority order would look like.
3. **Hierarchical** (Genesis 10). Ham and Shem are each latitudinally ordered on their own, in opposite directions, and cancel once concatenated; the Japheth branch shows only a weak trend in the same framework. A genealogy assembled from older regional lists would behave this way.

Fargard 1 fits none of the three; no test returns a robust geographic signal for it. It is also the one corpus here whose framing is cosmological rather than geographic or administrative—it lists the lands Ahura Mazdā created, each set against a counter-creation of Anra Mainyu—and a list organised by creational sequence has no reason to track geography. Read beside Bīsutūn, which names much the same world and *is* ordered, Fargard 1 sharpens the suggestion that genre, not language family, is what matters. We state this as a contrast, not a controlled result: the two texts are not matched on date.

Why genre should matter, we cannot say from five texts. Three factors would need to be separated in a larger sample: genre proper (an invoked hymn, a canonical political list, an imperial declaration, a cosmology, a genealogy); date, together with the depth of oral transmission behind the written text; and how linear the composer’s own working geography was likely to be—someone whose world runs along a single dominant axis may default to it.

5.2 What it does not imply

A result here says nothing about whether any tradition had special geographic-cognitive access, nothing about the identity, ethnicity, or migration history of any composer, and nothing for or against any particular philological identification beyond what published scholarship already supports. The tests bear only on whether the listed order correlates with the modern coordinates of the identified items.

5.3 Limitations

The most serious limitation is the small number of corpora. Five texts differing at once in length, identification quality, genre, and date leave those factors hopelessly confounded; the genre reading of §5.1 is a hypothesis the present data suggest, not one they establish. Isolating any single factor would need a sample several times larger.

Two measurement choices also bear on the result. A centroid is a coarse summary of a long linear feature, whether a river or a province; for the Mahājanapadas and the broader Genesis 10 regions, choosing the capital city instead of the administrative-area mean shifts individual coordinates by tens of kilometres. We checked that the cross-corpus comparison survives that choice, but it is a commitment, not a fact of the data. And the Avesta and Genesis 10 corpora carry a non-trivial share of lower-confidence identifications; dropping them leaves the qualitative conclusions in place, but a full robustness study of the identifications is a separate piece of work.

Finally, the Mallows model behind the likelihood ratio is one defensible choice among several. A Cayley- or Hamming-distance Mallows, or a non-Mallows model, could move the numbers. We use the Kendall-distance form because it is the one that matches Kendall’s τ as the test statistic.

5.4 Extensions

The same method runs on any fixed-order enumeration with identifiable items. Obvious candidates for a larger study are the Mahābhārata Bhīṣmaparvan river list, Strabo’s enumeration of Asian peoples in books 11–15, the directional mountain sequences of the Shan Hai Jing, and the legendary regions of Avesta Yašt 19. A comparative set of twenty or thirty such lists would begin to let the genre, length, and date factors be separated. Two refinements of the method itself would also help: distance-matrix correlations beyond the single Mantel statistic used here, and constrained nulls whose prior is built from attested intertextual list-traditions rather than from metre alone.

6 Conclusion

Geographic ordering is not a universal property of premodern lists. Of the five enumerations tested, the Ṛgveda Nadīstuti and the Anguttara Mahājanapadas trace a strong monotonic line along the region’s east-to-west axis ($|\tau| > 0.78$, $p < 10^{-4}$); the Bīsutūn dahyāva is ordered more loosely, by satrapal bloc; Genesis 10 is ordered within two of its three sub-sections but not as a flat sequence; and Vendīdād Fargard 1 shows no robust geographic order. Across the five, genre appears to track these outcomes more closely than language family or date does; this is

the paper’s most exploratory claim, and a larger comparative sample is needed to settle it. The method carries over to any other fixed-order enumeration of identifiable places.

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Disclosure of AI assistance

This manuscript and its accompanying analysis code were prepared with the assistance of a large language model (Anthropic’s Claude). The model assisted in writing and debugging the analysis pipeline, in assembling the corpus coordinate tables from the cited scholarly sources (every identification is attributed to a published source in the deposited CSVs), and in drafting and editing the prose. All statistical results were verified by independent re-execution of the pipeline (the replication repository includes a script that checks every reported number); all references were checked against authoritative bibliographic databases; and the author reviewed all content, made all substantive decisions, and takes full responsibility for the manuscript.

Data and code availability

All data, code, cached Overpass responses, and the random seed (20260518) used for the permutation tests are available at (Kulkarni, 2026); the exact revision behind every number in this paper is tagged **paper-v1** in that repository. Each corpus’s coordinate CSV, including identification provenance and confidence levels, is provided in **corpora/**. Reproduction instructions are given in the repository’s README and the **reproduce.sh** script.

A The Rgveda 10.75.5 corpus in detail

RV 10.75.5 has the strongest geographic signal of the five corpora, and the longest scholarly history—it has been read as evidence for Vedic-period geography since Roth (1846) and Zimmer (1879). This appendix sets out the data and the sensitivity analyses behind the statement in Section 3.1 that the result is robust.

The verse. RV 10.75.5 names its ten rivers in a single ṛc, here in the Aufrecht text (Aufrecht, 1877), checked against the metrically restored text of van Nooten and Holland (1994):

*imaṃ me gaṇge yamune sarasvatī
śutudri stomam sacatā paruṣṇyā
asiknyā marudvṛdhe vitastayā
ārjikiye śṛṇuhyā suṣomayā* (RV 10.75.5)

A note on one pair: the standard text places Asiknī (Chenab) at position 6 and Marudvṛdhā at position 7, directly after Paruṣṇī (Ravi). English translations sometimes invert the two, because *asiknyā* is instrumental and the English word order departs from the Sanskrit; the Sanskrit sequence is not in question. The analysis follows it.

Identifications and coordinates. Table 2 lists the ten rivers, their mainstream identifications (Witzel, 1995; Macdonell and Keith, 1912), and the centroid coordinates used. Two of the ten are not settled. The referent of Ārjikiyā is genuinely unsettled: the Vipās (Beas) is the identification of the older commentarial tradition, the Haro a more recent proposal, and other readings place it elsewhere in the upper Indus drainage (Witzel, 1999). The Haro coordinates are used for the primary analysis; the sensitivity analysis below spans six variant identifications, including readings well to the east of the Haro, and τ stays strongly negative throughout. The historical course of the Sarasvatī is likewise a long-running question, carried through the same sensitivity

Table 2: The ten rivers of RV 10.75.5 in verse order, with mainstream identifications and centroid coordinates. “ n_{nodes} ” is the number of OSM geometry nodes behind the centroid; Marudvṛdhā is hand-curated.

Pos.	Vedic name	Identified river	Lon (° E)	Lat (° N)	n_{nodes}
1	Gaṅgā	Ganges	81.88	26.87	15 159
2	Yamunā	Yamuna	78.09	29.10	19 337
3	Sarasvatī	Ghaggar–Hakra	75.76	29.87	12 673
4	Śutudrī	Sutlej	77.09	31.30	4 047
5	Paruṣṇī	Ravi	74.22	31.48	12 047
6	Asiknī	Chenab	73.68	31.97	12 824
7	Marudvṛdhā	Chenab–Jhelum doab (uncertain)	73.10	32.20	—
8	Vitastā	Jhelum	73.51	33.27	13 624
9	Ārjīkīyā	Haro (contested)	72.84	33.86	2 018
10	Suṣomā	Soan (Sohan/Sawan)	72.75	33.37	3 190

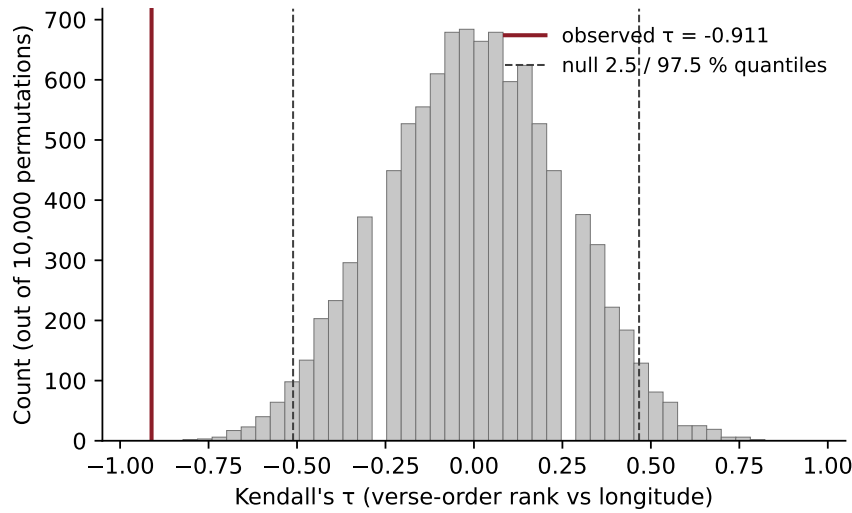


Figure 2: Null distribution of Kendall’s τ between verse-order rank and longitude, over 10 000 random permutations of the verse-order labels. The observed $\tau = -0.911$ falls far outside the central 95 % interval of the null.

analysis. Nine centroids were extracted from OpenStreetMap; Marudvṛdhā, for which no OSM-tagged river of that name exists, is hand-curated to the Chenab–Jhelum doab and is likewise exercised in the sensitivity analysis.

Primary test. Kendall’s τ between verse order and longitude is $\tau = -0.911$. Because $n = 10$, the permutation null can be replaced by the exact Kendall- τ distribution: of the $10! = 3,628,800$ orderings of ten items, exactly 108 reach $|\tau| \geq 0.911$, an exact two-sided $p = 2.98 \times 10^{-5}$. The 10 000-draw permutation null reaches no statistic that extreme and so reports only its resolution floor, $p < 10^{-4}$; its distribution (Figure 2) is near-symmetric about zero, mean -0.001 , standard deviation 0.251, central 95 % interval $[-0.511, +0.467]$.

Sensitivity analyses. Four checks are summarised in Table 3.

Geographic axis. τ against latitude is $+0.956$ (exact two-sided $p = 5.51 \times 10^{-6}$) and against the first principal component $+0.911$ (exact $p = 2.98 \times 10^{-5}$, equal to the longitude axis since the τ magnitude is the same). The latitude figure is not independent corroboration, since the rivers lie on a NW–SE diagonal and the two axes are strongly correlated; it shows only that the result is not an artefact of choosing longitude.

Table 3: RV 10.75.5 sensitivity-analysis summary. Every variant has $n \leq 10$, so “ p (two-sided)” is the *exact* two-sided Kendall- τ p -value rather than a permutation estimate.

Analysis	τ	p (two-sided)	Note
Primary: longitude	−0.911	2.98×10^{-5}	modern centroids
Axis: latitude	+0.956	5.51×10^{-6}	rivers also trend N-S
Axis: PC1	+0.911	2.98×10^{-5}	principal component of (lon, lat)
Paleocourse substitution	−0.778	9.46×10^{-4}	paleo-Sarasvatī, paleo-Sutlej
ID: Ārj.=Haro; Mar.=Chenab–Jhelum doab	−0.911	2.98×10^{-5}	
ID: Ārj.=Haro; Mar.=eastern alternative	−0.911	2.98×10^{-5}	
ID: Ārj.=Soan; Mar.=Chenab–Jhelum doab	−0.689	0.0047	
ID: Ārj.=Soan; Mar.=eastern alternative	−0.689	0.0047	
ID: Ārj.=upper Indus tributary; Mar.=Chenab–Jhelum doab	−0.867	1.15×10^{-4}	
ID: Ārj.=upper Indus tributary; Mar.=eastern alternative	−0.867	1.15×10^{-4}	
ID: Ārj.=Haro; Mar.=DROPPED	−0.944	4.96×10^{-5}	
Coord. perturb. ($\sigma = 0.5^\circ$, $n=1000$)	−0.822	—	95% CI [−0.956, −0.644]

Paleocourse. We re-run the test with published paleocourse centroids in place of the modern ones for two rivers: the Ghaggar–Hakra reconstruction of Giosan et al. (2012) and Clift et al. (2012) for the Sarasvatī, whose course is still under investigation (Singh et al., 2017), and the older pre-capture course of Yashpal et al. (1980) for the Śutudrī. This gives $\tau = -0.778$ (exact $p = 9.46 \times 10^{-4}$), attenuated but still clear.

Alternative identifications. Across six variant identifications of the disputed Ārjīkīyā and Marudvṛdhā, τ stays between −0.689 and −0.911, all $p \leq 0.005$. Dropping the contested Marudvṛdhā entirely ($n = 9$) gives $\tau = -0.944$.

Coordinate perturbation. Gaussian noise at $\sigma = 0.5^\circ$, over 1 000 iterations, yields a 95 % interval for τ of [−0.956, −0.644], median −0.822.

Every variant leaves the result in the same place: the order of rivers in RV 10.75.5 is strongly aligned with longitude under every null model tested.

References

- E. Vernon Arnold. *Vedic Metre in its Historical Development*. Cambridge University Press, Cambridge, 1905. Reprinted, Motilal Banarsidass, Delhi, 1967.
- Theodor Aufrecht. *Die Hymnen des Ṛigveda*. Adolph Marcus, Bonn, second edition, 1877. Standard reference edition of the Ṛgveda.
- Pierre Briant. *From Cyrus to Alexander: A History of the Persian Empire*. Eisenbrauns, Winona Lake, IN, 2002. Translated by Peter T. Daniels.
- Johannes Bronkhorst. *Greater Magadha: Studies in the Culture of Early India*. Brill, Leiden, 2007.
- Peter D. Clift, Andrew Carter, Liviu Giosan, Julie Durcan, Geoff A. T. Duller, Mark G. Macklin, Anwar Alizai, Ali R. Tabrez, Mohammad Danish, Sam VanLaningham, and Dorian Q. Fuller. U-Pb zircon dating evidence for a Pleistocene Sarasvati River and capture of the Yamuna River. *Geology*, 40(3):211–214, 2012. doi: 10.1130/G32840.1.
- Michael A. Fligner and Joseph S. Verducci. Distance based ranking models. *Journal of the Royal Statistical Society B*, 48(3):359–369, 1986. doi: 10.1111/j.2517-6161.1986.tb01420.x.
- Liviu Giosan, Peter D. Clift, Mark G. Macklin, Dorian Q. Fuller, Stefan Constantinescu, Julie A. Durcan, Thomas Stevens, Geoff A. T. Duller, Ali R. Tabrez, Kavita Gangal, Ronjoy Adhikari, Anwar Alizai, Florin Filip, Sam VanLaningham, and James P. M. Syvitski. Fluvial landscapes of the Harappan civilization. *Proceedings of the National Academy of Sciences*, 109(26):E1688–E1694, 2012. doi: 10.1073/pnas.1112743109.

- Gherardo Gnoli. *Zoroaster's Time and Homeland: A Study on the Origins of Mazdeism and Related Problems*. Number 7 in Series Minor. Istituto Universitario Orientale, Naples, 1980.
- John C. Gower. Generalized Procrustes analysis. *Psychometrika*, 40(1):33–51, 1975. doi: 10.1007/BF02291478.
- Richard S. Hess. *Studies in the Personal Names of Genesis 1–11*, volume 234 of *Alter Orient und Altes Testament*. Butzon & Bercker; Neukirchener Verlag, Kevelaer and Neukirchen-Vluyn, 1993.
- Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995. doi: 10.1080/01621459.1995.10476572.
- Maurice G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938. doi: 10.1093/biomet/30.1-2.81.
- Roland G. Kent. *Old Persian: Grammar, Texts, Lexicon*. American Oriental Society, New Haven, second edition, 1953.
- Ajinkya Kulkarni. Replication materials for “Geographic encoding in premodern enumerations: a five-corpus quantitative test”. <https://github.com/ajinkyakulkarni/geographic-encoding-enumerations>, 2026.
- Bimala Churn Law. *Tribes in Ancient India*. Bhandarkar Oriental Research Institute, Poona, second edition, 1973.
- Arthur A. Macdonell and Arthur B. Keith. *Vedic Index of Names and Subjects*, volume 2. John Murray, London, 1912.
- C. L. Mallows. Non-null ranking models. I. *Biometrika*, 44(1/2):114–130, 1957. doi: 10.1093/biomet/44.1-2.114.
- Nathan Mantel. The detection of disease clustering and a generalized regression approach. *Cancer Research*, 27(2):209–220, 1967.
- Roger Newson. Parameters behind “nonparametric” statistics: Kendall’s tau, Somers’ D and median differences. *Stata Journal*, 2(1):45–64, 2002. doi: 10.1177/1536867X0200200103.
- Pedro R. Peres-Neto and Donald A. Jackson. How well do multivariate data sets match? The advantages of a Procrustean superimposition approach over the Mantel test. *Oecologia*, 129(2):169–178, 2001. doi: 10.1007/s004420100720.
- T. W. Rhys Davids. *Buddhist India*. T. Fisher Unwin, London, 1903.
- Rudolf Roth. *Zur Litteratur und Geschichte des Weda*. A. Liesching & Comp., Stuttgart, 1846.
- Rüdiger Schmitt. *Wörterbuch der altpersischen Königsinschriften*. Reichert Verlag, Wiesbaden, 2014.
- Ajit Singh, Kristina J. Thomsen, Rajiv Sinha, Jan-Pieter Buylaert, Andrew Carter, Darren F. Mark, Philippa J. Mason, Alexander L. Densmore, Andrew S. Murray, Mayank Jain, Debajyoti Paul, and Sanjeev Gupta. Counter-intuitive influence of Himalayan river morphodynamics on Indus Civilisation urban settlements. *Nature Communications*, 8:1617, 2017. doi: 10.1038/s41467-017-01643-9.
- Barend A. van Nooten and Gary B. Holland. *Rig Veda: A Metrically Restored Text with an Introduction and Notes*, volume 50 of *Harvard Oriental Series*. Harvard University Press, Cambridge, MA, 1994.

- Gordon J. Wenham. *Genesis 1–15*. Word Biblical Commentary. Word Books, Waco, TX, 1987.
- Claus Westermann. *Genesis 1–11: A Commentary*. Augsburg, Minneapolis, 1984. Translated by John J. Scullion.
- Michael Witzel. Ṛgvedic history: Poets, chieftains and polities. In George Erdosy, editor, *The Indo-Aryans of Ancient South Asia: Language, Material Culture and Ethnicity*, volume 1 of *Indian Philology and South Asian Studies*, pages 307–352. Walter de Gruyter, Berlin and New York, 1995.
- Michael Witzel. Substrate languages in Old Indo-Aryan (Ṛgvedic, Middle and Late Vedic). *Electronic Journal of Vedic Studies*, 5(1):1–67, 1999.
- Michael Witzel. The home of the Aryans. In Almut Hintze and Eva Tichy, editors, *Anūsantatyai: Festschrift für Johanna Narten zum 70. Geburtstag*, number 19 in *Münchener Studien zur Sprachwissenschaft, Beiheft*, pages 283–338. J. H. Röll Verlag, Dettelbach, 2000.
- Dai Yamazaki, Daiki Ikeshima, Jeison Sosa, Paul D. Bates, George H. Allen, and Tamlin M. Pavelsky. MERIT Hydro: A high-resolution global hydrography map based on latest topography dataset. *Water Resources Research*, 55(6):5053–5073, 2019. doi: 10.1029/2019WR024873.
- Yashpal, Baldev Sahai, R. K. Sood, and D. P. Agrawal. Remote sensing of the “lost” Sarasvati river. *Proceedings of the Indian Academy of Sciences (Earth and Planetary Sciences)*, 89(3): 317–331, 1980.
- Heinrich Zimmer. *Altindisches Leben*. Weidmann, Berlin, 1879.